

Lecture 4: Interpolation via the Vandermonde System

CSE 330 - Numerical Methods (Class Notes)

1 Basics: The Vandermonde Method

To interpolate $n + 1$ data points $(x_0, y_0), \dots, (x_n, y_n)$ with a polynomial of degree n ,

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n,$$

we force the polynomial through every point. Writing $P_n(x_i) = y_i$ for all i gives the linear system

$$\underbrace{\begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix}}_{\text{Vandermonde matrix } V} \underbrace{\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix}}_{\text{coefficients } a} = \underbrace{\begin{pmatrix} f(x_0) \\ f(x_1) \\ \vdots \\ f(x_n) \end{pmatrix}}_f.$$

$$V \cdot a = f \quad \implies \quad a = V^{-1} \cdot f.$$

2 Worked Example (Age vs. Weight)

x (Age)	y (Weight)
10	40
30	60

With 2 data points we use a degree-1 polynomial ($n + 1$ points $\Rightarrow$ degree n), so the x^2 term is dropped:

$$P(x) = a_0 + a_1x.$$

Plugging in the points:

$$a_0 + 10a_1 = 40, \quad a_0 + 30a_1 = 60.$$

Vandermonde matrix:

$$\begin{pmatrix} 1 & 10 \\ 1 & 30 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 40 \\ 60 \end{pmatrix}.$$

Solve by inverting:

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 1 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 40 \\ 60 \end{bmatrix} = \frac{1}{30 - 10} \begin{bmatrix} 30 & -10 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 40 \\ 60 \end{bmatrix}$$

$$= \frac{1}{20} \begin{bmatrix} 600 \\ 20 \end{bmatrix} = \begin{bmatrix} 30 \\ 1 \end{bmatrix}.$$

Hence $a_0 = 30$, $a_1 = 1$, giving

$$\boxed{P(x) = 30 + x}$$

3 Worked Example

We are given the following data:

x (radians)	y
$-\frac{\pi}{2}$	3
$\frac{\pi}{4}$	4.5
$\frac{3\pi}{4}$	6
π	6.5

Note (radian mode). The x -values are angles measured in *radians*, so all numerical work uses their radian decimal values (with $\pi \approx 3.14$), *not* degrees. For example $x_0 = -\pi/2 \approx -1.57$ (a pure number), not -90 .

There are 4 data points, so $n = 3$. We must find the cubic interpolating polynomial

$$P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

using the Vandermonde method, i.e. determine the coefficient vector $a = (a_0, a_1, a_2, a_3)^T$ from $a = V^{-1}f$.

4 Step 1 - Evaluate the Nodes

Numerically (radian mode, $\pi \approx 3.14$), the four nodes are

$$\begin{aligned} x_0 &= -\frac{\pi}{2} \approx -1.57, & x_1 &= \frac{\pi}{4} \approx 0.79, \\ x_2 &= \frac{3\pi}{4} \approx 2.36, & x_3 &= \pi \approx 3.14. \end{aligned}$$

Now force the cubic through each point, $P_3(x_i) = y_i$, in the form $a_0 + a_1x_i + a_2x_i^2 + a_3x_i^3 = y_i$. Substituting the radian values (powers rounded to two decimals) gives one equation per node:

$$\begin{aligned} P_3(x_0) &= a_0 + a_1(-1.57) + a_2(2.47) + a_3(-3.88) = 3, \\ P_3(x_1) &= a_0 + a_1(0.79) + a_2(0.62) + a_3(0.48) = 4.5, \\ P_3(x_2) &= a_0 + a_1(2.36) + a_2(5.55) + a_3(13.08) = 6, \\ P_3(x_3) &= a_0 + a_1(3.14) + a_2(9.87) + a_3(31.01) = 6.5. \end{aligned}$$

5 Step 2 - Build the V , a , and f Matrices

$$V = \begin{pmatrix} 1 & x_0 & x_0^2 & x_0^3 \\ 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 1 & x_3 & x_3^2 & x_3^3 \end{pmatrix} = \begin{pmatrix} 1 & -1.57 & 2.47 & -3.88 \\ 1 & 0.79 & 0.62 & 0.48 \\ 1 & 2.36 & 5.55 & 13.08 \\ 1 & 3.14 & 9.87 & 31.01 \end{pmatrix}$$

$$a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad f = \begin{pmatrix} 3 \\ 4.5 \\ 6 \\ 6.5 \end{pmatrix}.$$

The system to solve is $V \cdot a = f$.

6 Step 3 - Invert V and Compute $a = V^{-1}f$

Finding V^{-1}

$V^{-1} = \frac{1}{\det V} \text{adj}(V)$ or, use calculator.

$$V = \begin{pmatrix} 1 & -1.57 & 2.47 & -3.88 \\ 1 & 0.79 & 0.62 & 0.48 \\ 1 & 2.36 & 5.55 & 13.08 \\ 1 & 3.14 & 9.87 & 31.01 \end{pmatrix}$$

$$V^{-1} = \begin{pmatrix} 0.13 & 1.34 & -0.81 & 0.34 \\ -0.27 & -0.14 & 0.76 & -0.35 \\ 0.14 & -0.45 & 0.49 & -0.18 \\ -0.02 & 0.11 & -0.21 & 0.11 \end{pmatrix}.$$

Computing $a = V^{-1}f$

We write the product $a = V^{-1}f$ as a matrix multiplication:

$$a = V^{-1}f = \begin{pmatrix} 0.13 & 1.34 & -0.81 & 0.34 \\ -0.27 & -0.14 & 0.76 & -0.35 \\ 0.14 & -0.45 & 0.49 & -0.18 \\ -0.02 & 0.11 & -0.21 & 0.11 \end{pmatrix} \begin{pmatrix} 3 \\ 4.5 \\ 6 \\ 6.5 \end{pmatrix}$$

Doing the row-by-column multiplication inside the matrix:

$$a = \begin{pmatrix} (0.13)(3) + (1.34)(4.5) + (-0.81)(6) + (0.34)(6.5) \\ (-0.27)(3) + (-0.14)(4.5) + (0.76)(6) + (-0.35)(6.5) \\ (0.14)(3) + (-0.45)(4.5) + (0.49)(6) + (-0.18)(6.5) \\ (-0.02)(3) + (0.11)(4.5) + (-0.21)(6) + (0.11)(6.5) \end{pmatrix}$$

which evaluates to the coefficient (a) matrix:

$$a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 3.77 \\ 0.85 \\ 0.17 \\ -0.11 \end{pmatrix}.$$

Coefficients:

$$a_0 = 3.77, \quad a_1 = 0.85, \quad a_2 = 0.17, \quad a_3 = -0.11.$$

7 Step 4 - Write the Polynomial $P_3(x)$

Substituting the coefficients into $P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$:

$$P_3(x) \approx 3.77 + 0.85x + 0.17x^2 - 0.11x^3$$